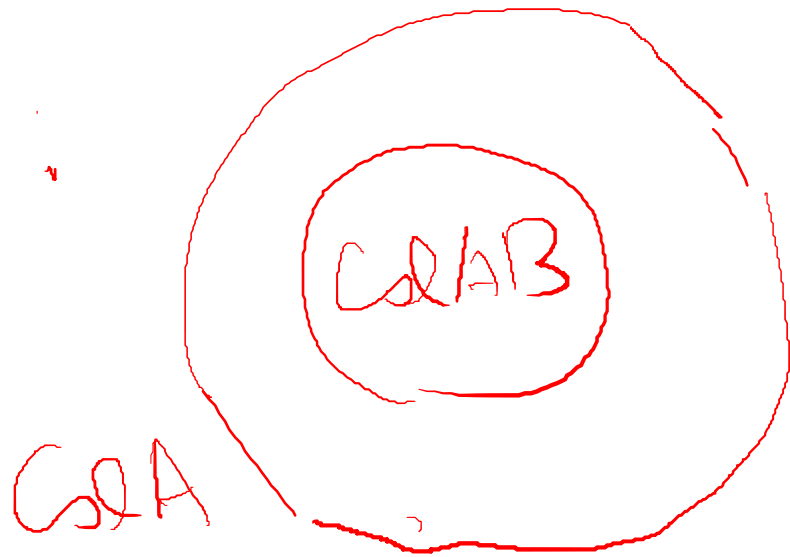


4-3

73. Prove that for any $m \times n$ matrix A and any $n \times p$ matrix B , the column space of AB is contained in the column space of A .

$$\text{Col } AB \subset \text{Col } A$$

$$w \in \text{Col } A \equiv w = Au$$



$$w \in \text{Col } AB$$

$$\Rightarrow w = A \underline{B u'}$$

$$\Rightarrow w = A u''$$

$$\Rightarrow w \in \text{Col } A$$

Hint: w in $\text{Col } AB \Rightarrow w = ABu$

$$w \in \text{Col } A \equiv w = Au$$

$$A = [a_1 \ a_2 \ \dots \ a_n]$$

$$w = c_1 \underline{a_1} + c_2 \underline{a_2} + \dots + c_n \underline{a_n}$$

$$w = [a_1 \ a_2 \ \dots \ a_n] \begin{matrix} \underbrace{\begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}}_u \end{matrix}$$

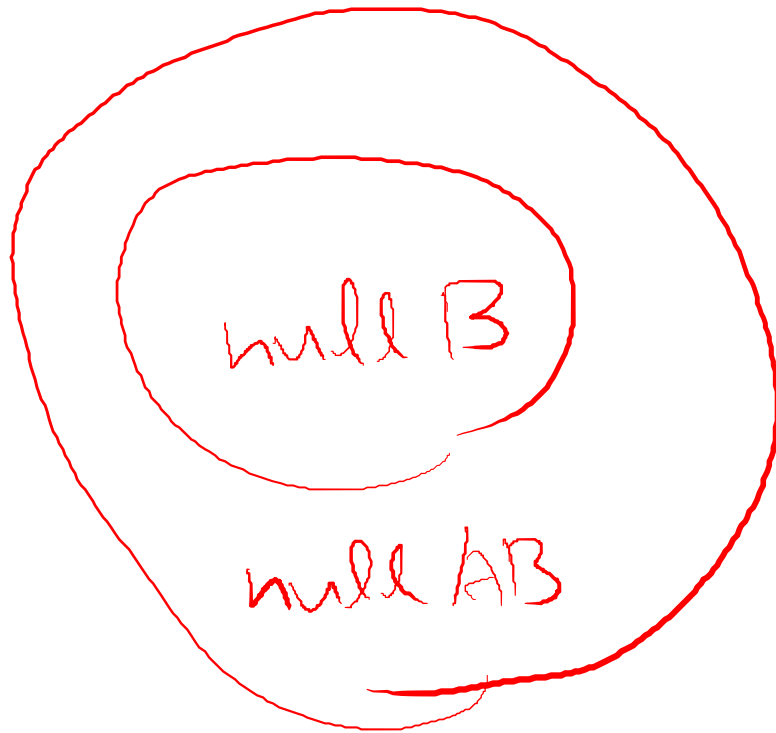
$$w \in \text{Col } AB$$

$$\Rightarrow w = A \boxed{Bu}$$

$$\Rightarrow w = A \underbrace{u'}_u$$

$$\Rightarrow w \in \text{Col } A$$

74. Prove that for any $m \times n$ matrix A and any $n \times p$ matrix B , the null space of B is contained in the null space of AB .



$$w \in \text{null } A \equiv 0 = Aw$$

$$w \in \text{null } B$$

$$\Rightarrow Bw = 0$$

$$\Rightarrow ABw = A0 = 0$$

$$\Rightarrow w \in \text{null } AB$$

75. Prove that for any $m \times n$ matrix A and any $n \times p$ matrix B , $\text{rank } AB \leq \text{rank } A$.

$$\underline{\dim(\text{Col } A)} = \underline{\text{rank } A}, \quad \underline{\text{Col } AB} \subset \underline{\text{Col } A}$$

$$\Rightarrow \underline{\dim(\text{Col } AB)} \leq \underline{\dim(\text{Col } A)}$$

$$\Rightarrow \text{rank } AB \leq \text{rank } A$$

76. Prove that for any $m \times n$ matrix A and any $n \times n$ matrix B , $\text{nullity} A \leq \text{nullity} AB$.

75. Prove that for any $m \times n$ matrix A and any $n \times p$ matrix B , $\text{rank} AB \leq \text{rank} A$.

77. Prove that for any $m \times n$ matrix A and any $n \times p$ matrix B , $\text{rank} AB \leq \text{rank} B$.

已知 $\text{rank} A = \text{rank} A^T$

Hint: The ranks of AB and $(AB)^T$ are equal.

$$\begin{aligned} \text{rank } AB &= \text{rank } (AB)^T \\ &= \text{rank } B^T A^T \leq \text{rank } B^T = \text{rank } B \end{aligned}$$

75. Prove that for any $m \times n$ matrix A and any $n \times p$ matrix B , $\text{rank} AB \leq \text{rank} A$.

78. Prove that for any $m \times n$ matrix A and any $n \times p$ matrix B , $\text{nullity } B \leq \text{nullity } AB$.

83. (a) Show that for any vector $\mathbf{u}$ in R^n , $\mathbf{u}^T \mathbf{u} = 0$ if and only if $\mathbf{u} = \mathbf{0}$.

$$\mathbf{u}^T \mathbf{u} = (u_1)^2 + \cdots + (u_n)^2$$

83. (b) Prove that for any matrix A , if $\mathbf{u}$ is in $RowA$ and $\mathbf{v}$ is in $NullA$, then $\mathbf{u}^T \mathbf{v} = 0$.

Hint: The row space of A equals the column space of A^T . $\mathbf{u} = A^T \mathbf{w}$

83. (c) Show that for any matrix A , if $\mathbf{u}$ belongs to both $RowA$ and $NullA$, then $\mathbf{u} = \mathbf{0}$.

84. Show that for any $m \times n$ matrix A , the union of a basis for $RowA$ and a basis for $NullA$ is a basis for R^n .

Strategy: (1) Show that the union of two bases has n vectors. (2) The n vectors are independent.

83. (c) Show that for any matrix A , if $\mathbf{u}$ belongs to both $RowA$ and $NullA$, then $\mathbf{u} = \mathbf{0}$.